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| **Subject:** | **Database Development 171/181** |
| **Assessment:** | **Assignment 1** |
| **Total:** | **60 Marks** |

# Assignment 1

## Introduction to Database Systems

Databases evolved from the need to manage large amounts of data in an organised and efficient manner. In the olden days, manual and then computerised file systems were used to organize such data. In a computerised file system, data is stored in many separate and unrelated files, resulting in what database professionals term islands of information. The problems inherent in file systems make using a database system very desirable. Databases represent a breakthrough in the way data is stored, accessed, and managed. A database management system (DBMS) provides numerous functionalities and eliminates most of the data management problems.

# Outline

You are asked to research on database systems and emerging data storage needs. Use sound reasoning and explanatory information for a reader without extensive technical knowledge of databases.

(a) A growing retail company stores its customers, sales, and inventory data in separate files on different computers.
   *(a)* Analyse the problems the business may face as it grows. *[6 marks]*
   (b) Recommend whether the company should implement a database and justify your answer. *[4 marks]*

(b) An online store has noticed customer complaints about incorrect deliveries and billing errors.
   *(a)* Discuss possible dimensions of data quality that could cause these problems. *[6 marks]*
   *(b)* Using practical examples, analyse how poor data quality can negatively affect business decision-making. *[4 marks]*

(c) Modern Relational DBMSs are much more than data storage systems but are a set of software components that perform various data management functions.
   *(a)* Explain why the DBMS is considered central to the functioning of the entire database system. *[5 marks]*
   *(b)* Describe at least five (5) major functions of a modern RDBMS. *[10 marks]*

(d) A new higher education institution requires a secure and scalable database to support its registration system.
   *(a)* Compare three different relational DBMS software *[6 marks]*
   *(b)* Recommend the most suitable option for the institution and give reasons for your choice. *[4 marks]*

(e) Explain the role of SQL in managing the new database system. In your answer describe specific tasks that can be performed using SQL. Include SQL code examples to illustrate your answer.

**Your presentation should not exceed 2000 words (about 4 pages single line spacing or 5 pages 1½ line spacing 11pt size font) excluding Cover Page, Table of Contents and Reference pages.**

## Mark Allocation

| Content | |
|---|---|
| The importance of a database | 10 |
| Dimensions and importance of data quality | 10 |
| The role and functions of a DBMS | 15 |
| Comparison of DBMS software | 10 |
| Use of Structured Query Language (SQL) | 10 |
| Organisation and Structure | 2 |
| Bibliography or List of References | 3 |
| **Total** | **60** |

## Additional Information

- This is an individual assignment. All work must be done on your own.
- Submit your assignment electronically on Moodle (BC Connect) before the due date.
- Include a cover page, table of contents and clear headings.
- All writing must be correctly referenced and cited.
- Belgium Campus makes use of software that can **scan for plagiarism** and penalties apply for any plagiarised work including but not limited to getting 0 for this assignment
- No mark will be awarded if the project is not uploaded via BC Connect assignment page.
- Late submissions will not be accepted and missing the deadline is an automatic **0**.